

Student Enrollment & Course Management System (SECMS)

Project Documentation

A Salesforce-Based Academic Enrollment Management Application

Submitted as a Group Project under the TN Skills - Naan Mudalvan Programme

S.No	Student Name	Registration Number	Contribution
1	Ruqaiya Sadeem K F	31224U09073	Developer account setup, Flow automation, Reports, Conclusion / documentation
2	Shaina Sadaf N Khan	31224U09081	Custom objects, fields and relationships
3	Moolakadai Musfera Konain	31224U09058	Tabs, Lightning App, Page Layouts and Record Types
4	Vinothini S	31224U09093	Validation Rules and Approval Process
5	Nivetha M	31224U09069	Dashboards and Security Setup

Group ID: SWTID-2026-4095

Bachelor of Computer Applications (BCA), 5th Semester
Islamiah Women's Arts and Science College, Vaniyambadi
Affiliated to Thiruvalluvar University
Academic Year 2026-2027

Project Overview and Scenarios

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Project Description

SECMS is a Salesforce-based academic management solution that brings student, course, instructor and enrollment information into one cloud environment. It uses declarative automation, validation rules, an approval process, reports, dashboards and security controls to support the enrollment lifecycle.

Objective

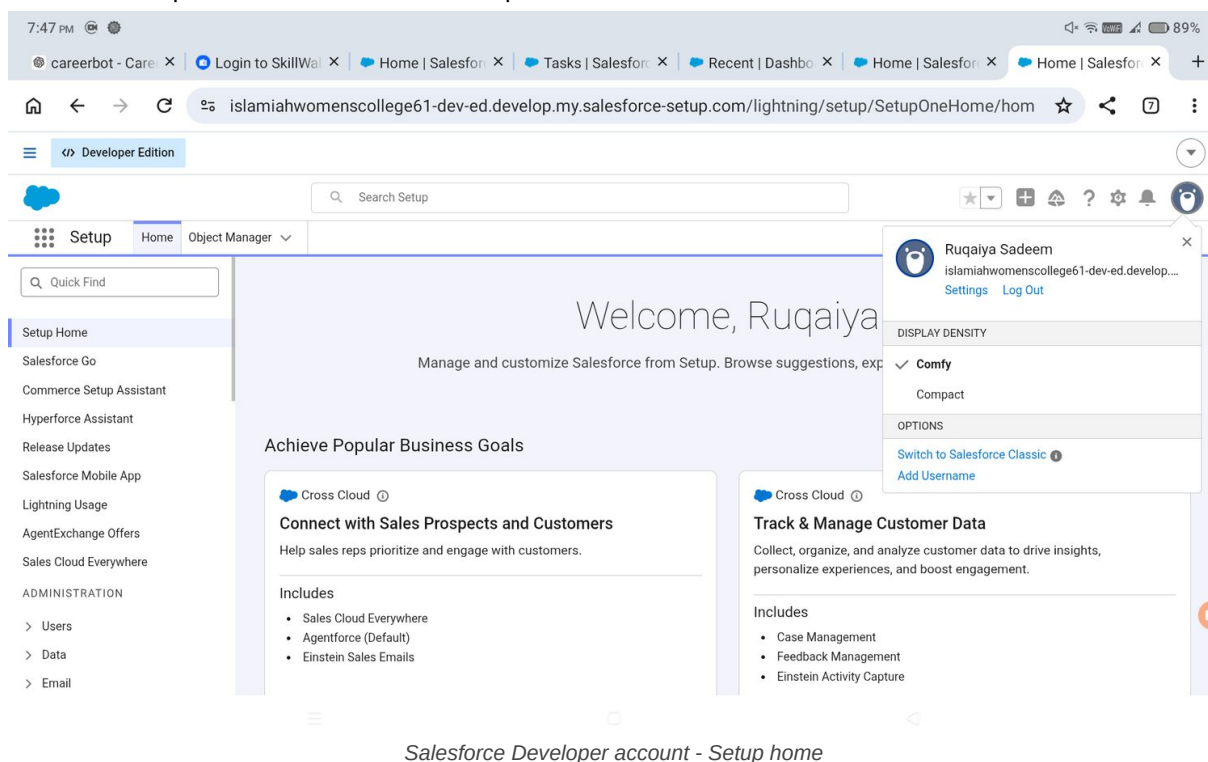
Replace manual and fragmented enrollment handling with a structured, scalable, automated and secure system that improves data consistency and makes enrollment status easier to track.

Scope

Data objects, fields, relationships, Lightning navigation, record types, page layouts, validation rules, approval processing, flows, reports, dashboards and security controls.

Key Scenarios

- Admin creates a student record with a valid email.
- Admin maintains courses and instructors.
- An enrollment is created and routed through automation and approval.
- Admin reviews reports and the dashboard for operational status.



Technical Architecture and Workflow

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Architecture

- Presentation: Salesforce Lightning App, tabs and record pages.
- Application / Automation: Flows, Approval Process and Validation Rules.
- Core Data: Student, Course, Instructor and Enrollment custom objects.
- Analytics: Reports and Student Management Dashboard.
- Security: Profiles, Permission Sets and Sharing / OWD.

Relationships

Enrollment is the central transaction object. It references Student, Course and Instructor through lookup relationships so the enrollment lifecycle can be viewed in context.

Project Workflow

- Identify user and process problems.
- Define the Student / Course / Instructor / Enrollment data model.
- Configure tabs, Lightning App, record types and layouts.
- Configure validation rules and build flows.
- Configure the approval process.
- Create reports and the dashboard.
- Configure security.
- Test and document the evidence.

Milestone 1 - Data Model: Objects

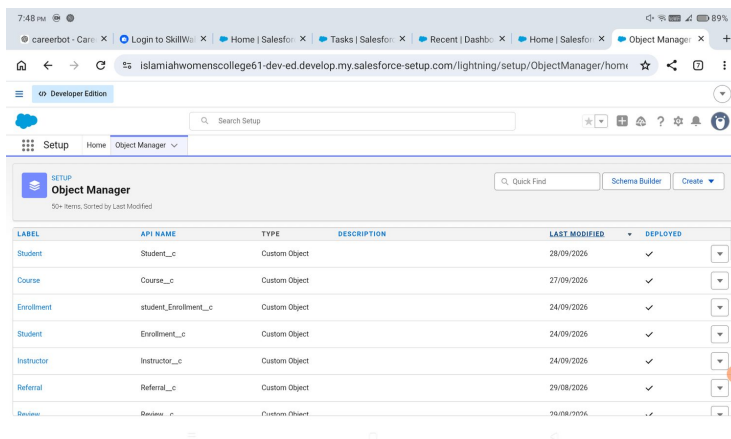
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Milestone 1 - Custom Objects

Four custom objects form the data layer: Student (personal and contact details, status), Course (category, duration, fee), Enrollment (transaction linking student, course and instructor) and Instructor (code, expertise, contact).

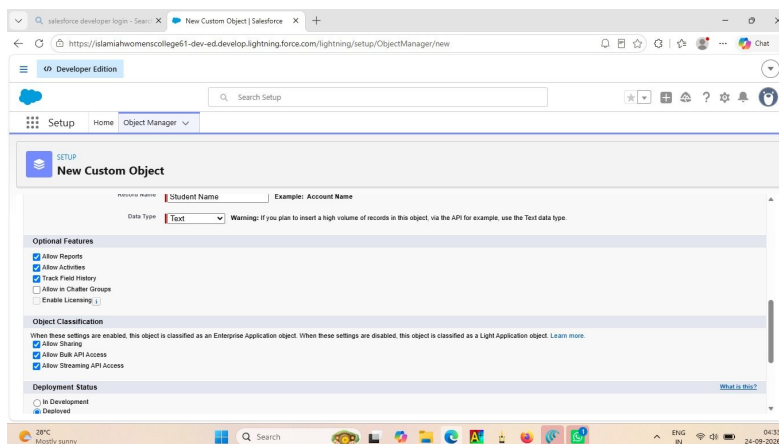
Development Process

- Open Setup > Object Manager > Create > Custom Object.
- Set label and plural label (e.g. Student / Students) and the record name.
- Enable reporting, activities and search options; save and verify in Object Manager.



LABEL	API NAME	TYPE	DESCRIPTION	LAST MODIFIED	DEPLOYED
Student	Student__c	Custom Object		28/09/2026	✓
Course	Course__c	Custom Object		27/09/2026	✓
Enrollment	student_Enrollment__c	Custom Object		24/09/2026	✓
Student	Enrollment__c	Custom Object		24/09/2026	✓
Instructor	Instructor__c	Custom Object		24/09/2026	✓
Referral	Referral__c	Custom Object		29/08/2026	✓
Reviews	Reviews__c	Custom Object		16/08/2026	✓

Object Manager showing the SECMS custom objects



NEW CUSTOM OBJECT

OBJECT NAME: Student Name | EXAMPLE: Account Name

DATA TYPE: Text | Warning: If you plan to insert a high volume of records in this object, via the API for example, use the Text data type

Optional Features

- ☒ Allow Reports
- ☒ Allow Activities
- ☒ Track Field History
- ☐ Allow in Chatter Group
- ☐ Enable Learning (1)

Object Classification

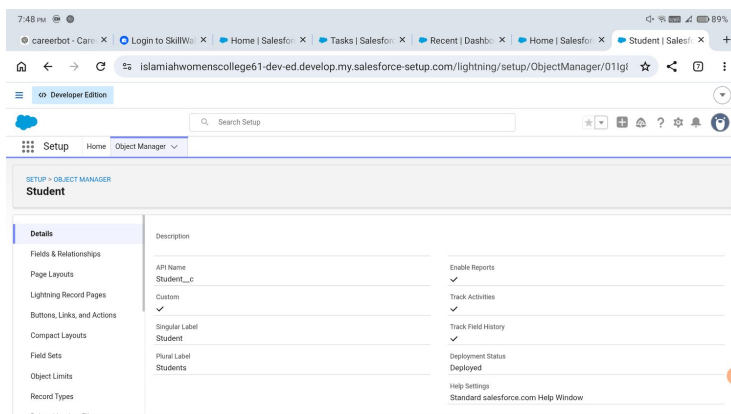
When these settings are enabled, this object is classified as an Enterprise Application object. When these settings are disabled, this object is classified as a Light Application object. [Learn more](#)

- ☒ Allow Sharing
- ☒ Allow Bulk API Access
- ☒ Allow Streaming API Access

Deployment Status

☐ In Development | ☒ Deployed

Creating the Student custom object



SETUP > OBJECT MANAGER

Student

Details

- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types

Description

API Name: Student__c

Custom

Singular Label: Student

Plural Label: Students

Enable Reports: ✓

Track Activities: ✓

Track Field History: ✓

Deployment Status: Deployed

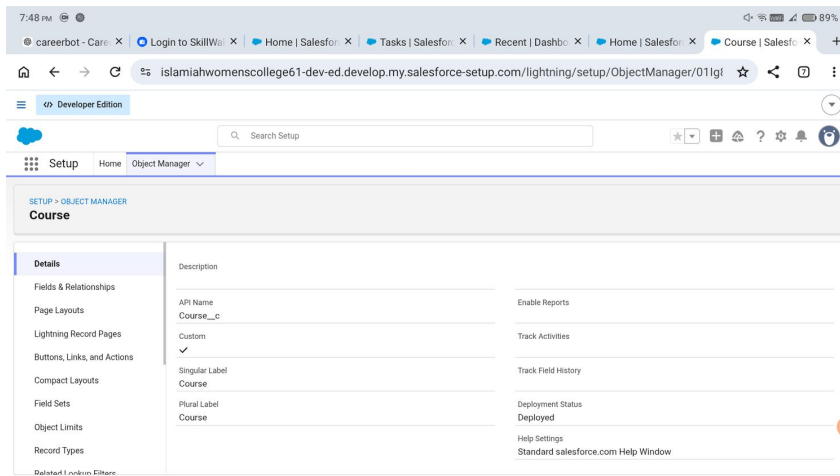
Help Settings: Standard Salesforce.com Help Window

Student custom object

Milestone 1 - Data Model: Objects (continued)

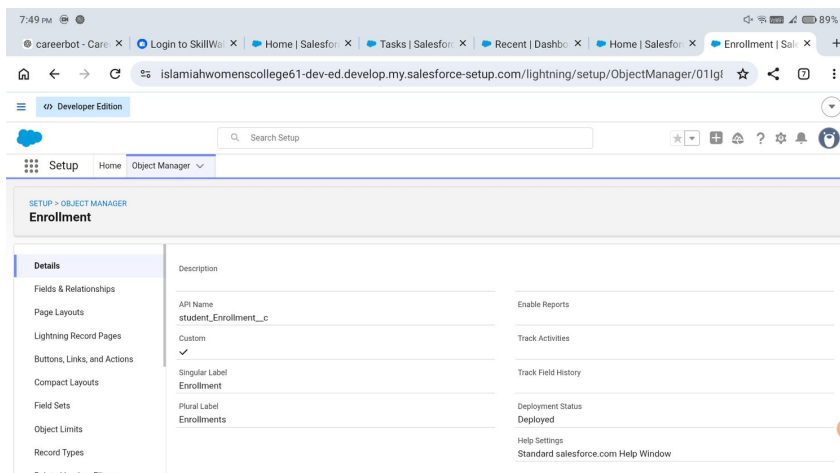
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The remaining object screens are shown below.



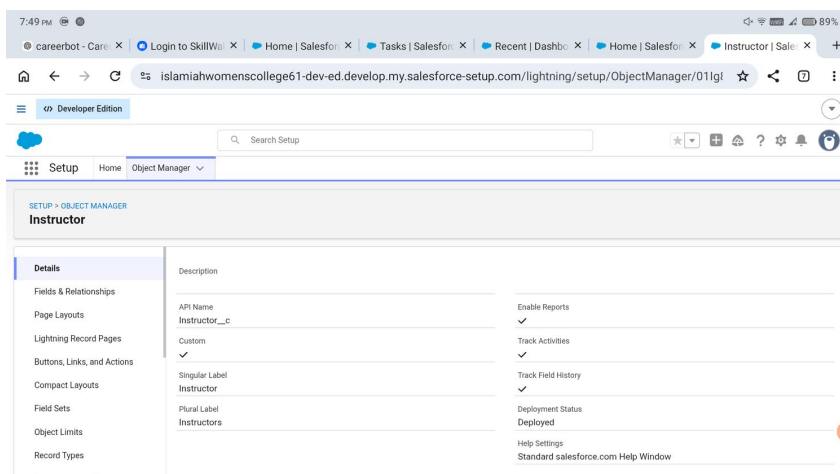
The screenshot shows the Salesforce Setup interface for the 'Course' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'SETUP > OBJECT MANAGER Course'. It includes a 'Description' field, an 'API Name' field with the value 'Course__c', and a 'Custom' checkbox that is checked. Below these are 'Singular Label' (Course) and 'Plural Label' (Course). On the right side, there are checkboxes for 'Enable Reports', 'Track Activities', 'Track Field History', and 'Deployment Status' (Deployed). A 'Help Settings' link is also present.

Course custom object



The screenshot shows the Salesforce Setup interface for the 'Enrollment' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'SETUP > OBJECT MANAGER Enrollment'. It includes a 'Description' field, an 'API Name' field with the value 'student_Enrollment__c', and a 'Custom' checkbox that is checked. Below these are 'Singular Label' (Enrollment) and 'Plural Label' (Enrollments). On the right side, there are checkboxes for 'Enable Reports', 'Track Activities', 'Track Field History', and 'Deployment Status' (Deployed). A 'Help Settings' link is also present.

Enrollment custom object



The screenshot shows the Salesforce Setup interface for the 'Instructor' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships, Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'SETUP > OBJECT MANAGER Instructor'. It includes a 'Description' field, an 'API Name' field with the value 'Instructor__c', and a 'Custom' checkbox that is checked. Below these are 'Singular Label' (Instructor) and 'Plural Label' (Instructors). On the right side, there are checkboxes for 'Enable Reports', 'Track Activities', 'Track Field History', and 'Deployment Status' (Deployed). A 'Help Settings' link is also present.

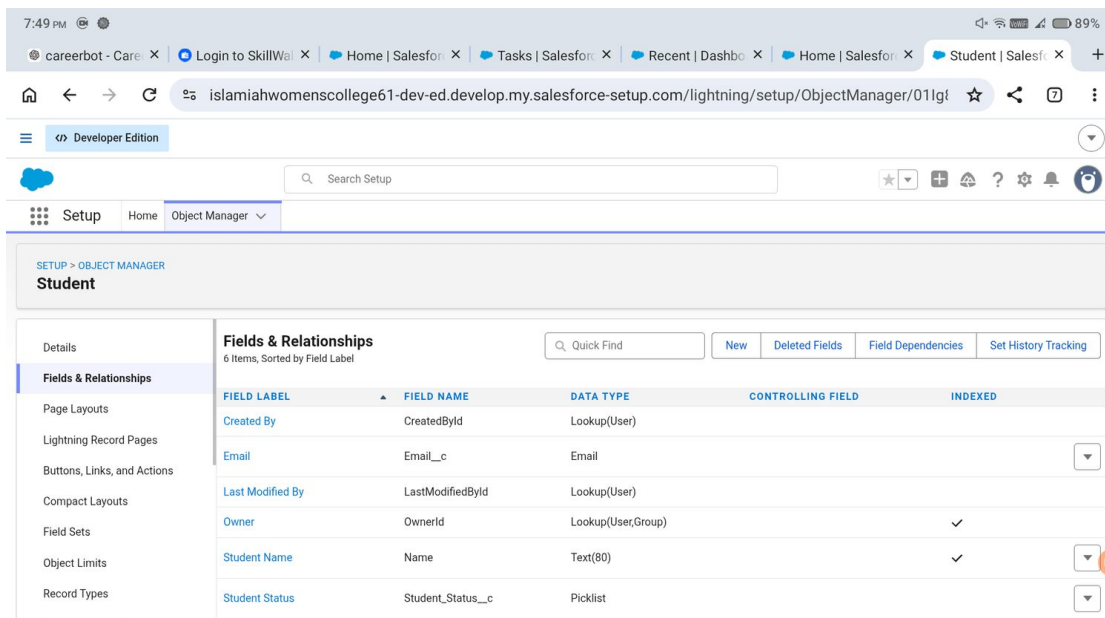
Instructor custom object

Milestone 2 - Fields

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Fields

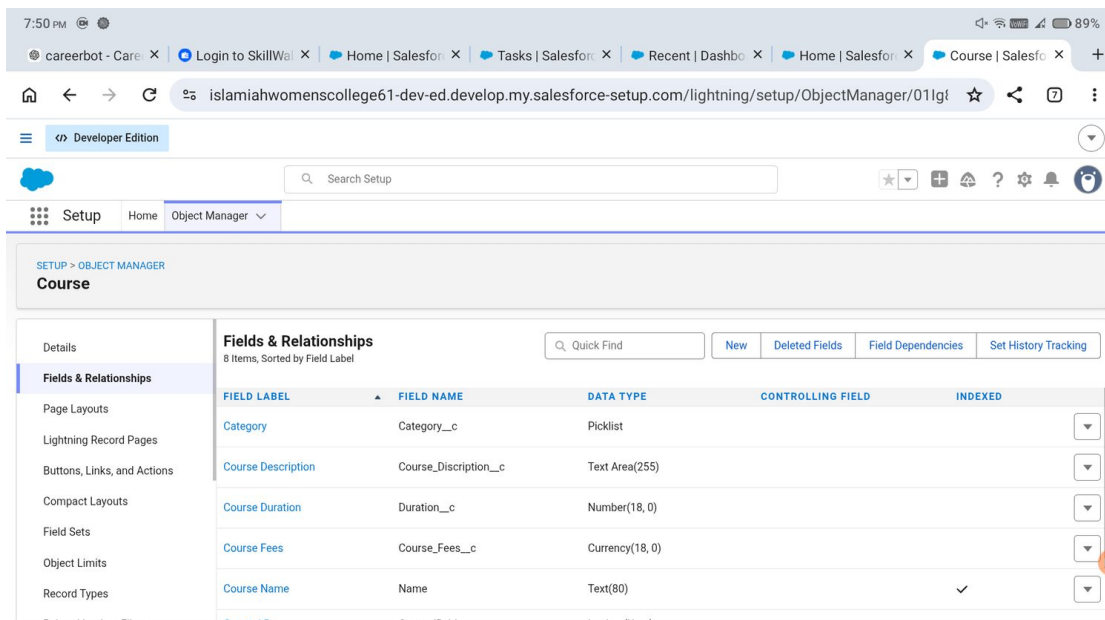
Student: Student Name, Email, Phone, Date of Birth, Student Status (picklist), Address. Course: Course Name, Category (picklist), Duration, Course Fee (currency), Description. Instructor: Instructor Name, Instructor Code, Expertise (picklist), Phone, Email. Enrollment: Student, Course and Instructor lookups, Enrollment Date, Enrollment Status (picklist), Fees Paid (checkbox), Total Amount (currency), Comments.



The screenshot shows the Salesforce Setup interface for the 'Student' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Student' and shows 'Fields & Relationships' with 6 items, sorted by Field Label. The table lists the following fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Last Modified By	LastModifiedById	Lookup(User)		
Owner	OwnerId	Lookup(User,Group)		✓
Student Name	Name	Text(80)		✓
Student Status	Student_Status__c	Picklist		

Student fields



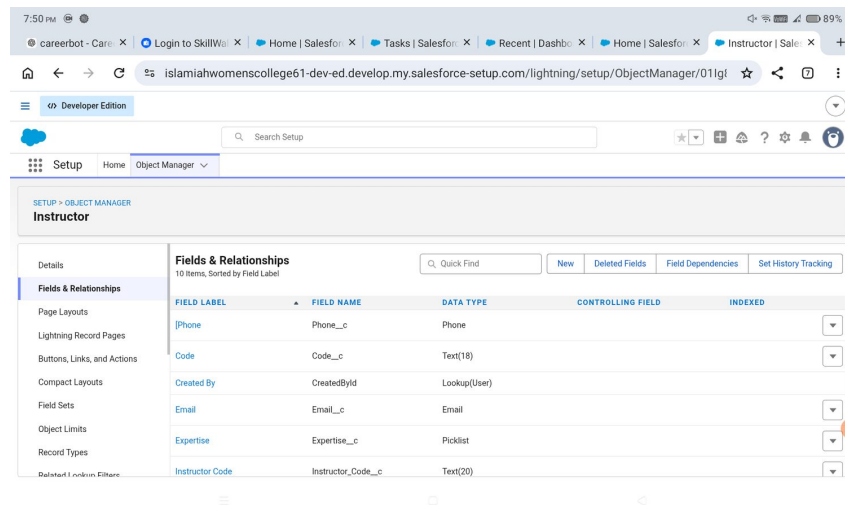
The screenshot shows the Salesforce Setup interface for the 'Course' object. The left sidebar contains a navigation menu with options: Details, Fields & Relationships (selected), Page Layouts, Lightning Record Pages, Buttons, Links, and Actions, Compact Layouts, Field Sets, Object Limits, and Record Types. The main content area is titled 'Course' and shows 'Fields & Relationships' with 8 items, sorted by Field Label. The table lists the following fields:

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Category	Category__c	Picklist		
Course Description	Course_Description__c	Text Area(255)		
Course Duration	Duration__c	Number(18, 0)		
Course Fees	Course_Fees__c	Currency(18, 0)		
Course Name	Name	Text(80)		✓
Created By	CreatedById	Lookup(User)		

Course fields

Milestone 2 - Fields (continued)

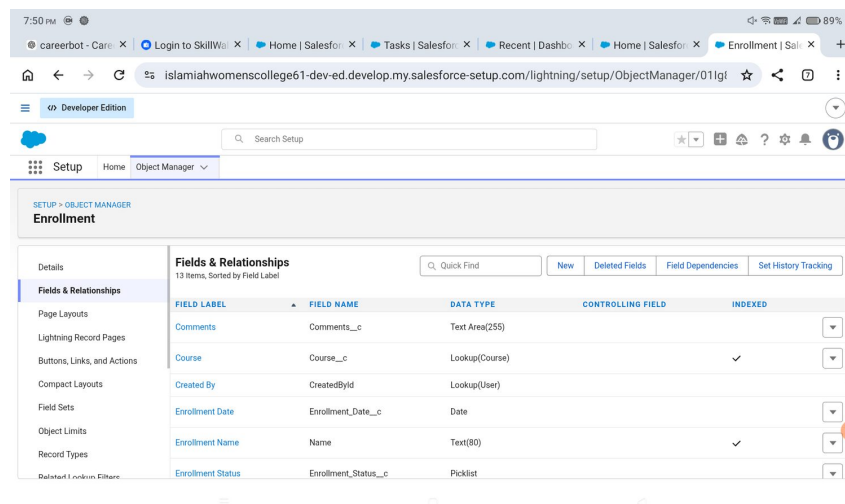
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The screenshot shows the Salesforce Setup interface for the 'Instructor' object. The 'Fields & Relationships' tab is selected, displaying a table of fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are Phone, Code, Created By, Email, Expertise, and Instructor Code.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Phone	Phone__c	Phone		
Code	Code__c	Text(18)		
Created By	CreatedById	Lookup(User)		
Email	Email__c	Email		
Expertise	Expertise__c	Picklist		
Instructor Code	Instructor_Code__c	Text(20)		

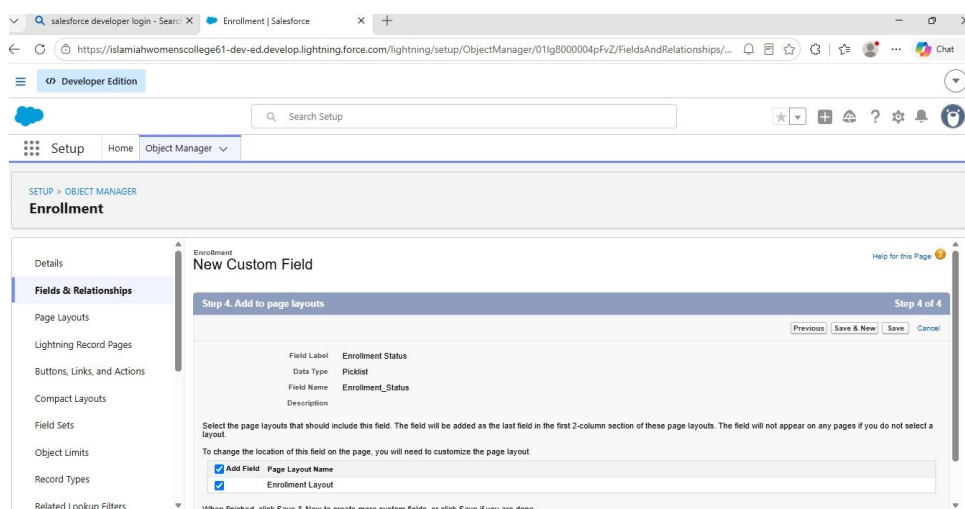
Instructor fields



The screenshot shows the Salesforce Setup interface for the 'Enrollment' object. The 'Fields & Relationships' tab is selected, displaying a table of fields. The table has columns for Field Label, Field Name, Data Type, Controlling Field, and Indexed. The fields listed are Comments, Course, Created By, Enrollment Date, Enrollment Name, and Enrollment Status.

FIELD LABEL	FIELD NAME	DATA TYPE	CONTROLLING FIELD	INDEXED
Comments	Comments__c	Text Area(255)		
Course	Course__c	Lookup(Course)		
Created By	CreatedById	Lookup(User)		
Enrollment Date	Enrollment_Date__c	Date		
Enrollment Name	Name	Text(80)		
Enrollment Status	Enrollment_Status__c	Picklist		

Enrollment fields and lookup relationships



The screenshot shows the Salesforce Setup interface for the 'Enrollment' object, specifically the 'New Custom Field' wizard. The wizard is at Step 4, 'Add to page layouts'. The field being created is 'Enrollment Status', with a data type of 'Picklist' and a field name of 'Enrollment_Status'. The wizard prompts the user to select the page layouts that should include this field. The 'Add Field' checkbox is checked, and the 'Page Layout Name' is 'Enrollment Layout'.

Field Label: Enrollment Status
Data Type: Picklist
Field Name: Enrollment_Status
Description:

Select the page layouts that should include this field. The field will be added as the last field in the first 2-column section of these page layouts. The field will not appear on any pages if you do not select a layout.

To change the location of this field on the page, you will need to customize the page layout.

☒ Add Field ☐ Page Layout Name
☒ Enrollment Layout

Creating the Enrollment Status picklist field

Milestone 2 - Relationships

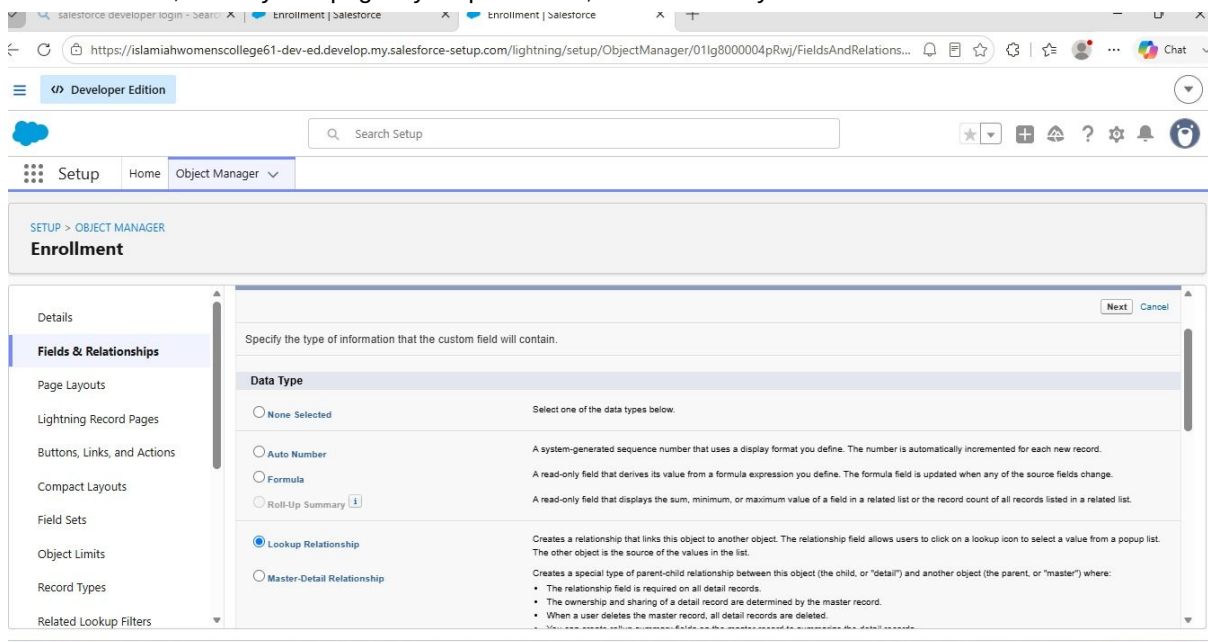
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Relationships

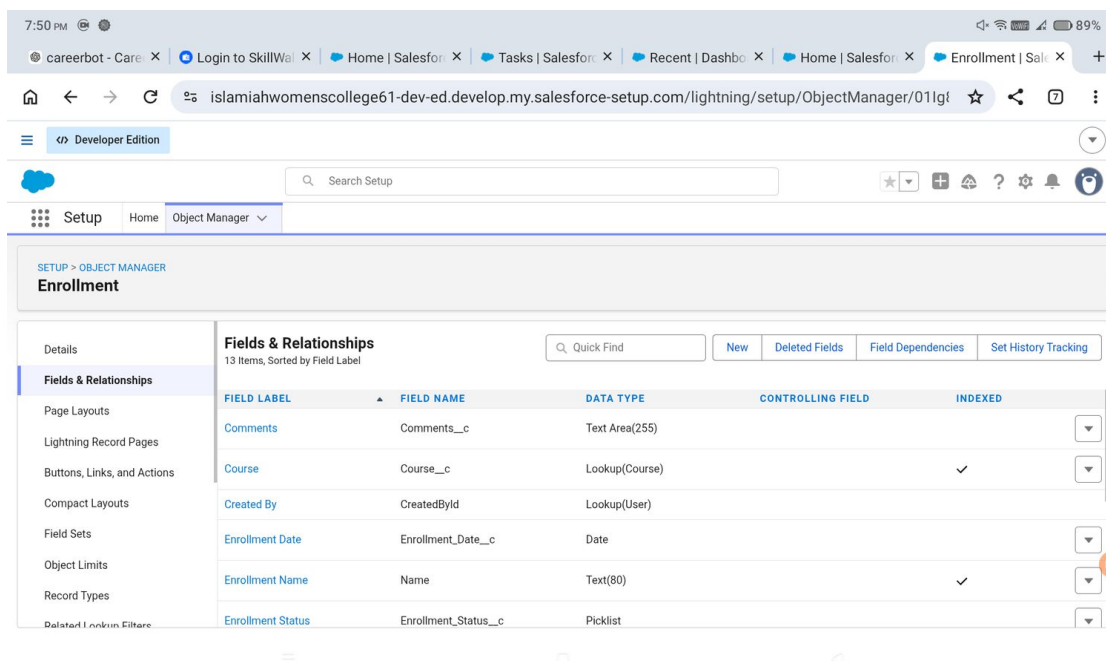
Lookup relationships from Enrollment to Student, Course and Instructor link each enrollment with its student, the selected course and the instructor. Sample relationship pattern: Student 1..* Enrollment *.1 Course; Enrollment *.1 Instructor.

Development Process

- Open Enrollment > Fields & Relationships > New.
- Choose Lookup Relationship and select the related object.
- Set the field label, visibility and page layout placement; save and verify.



Selecting Lookup Relationship data type on Enrollment



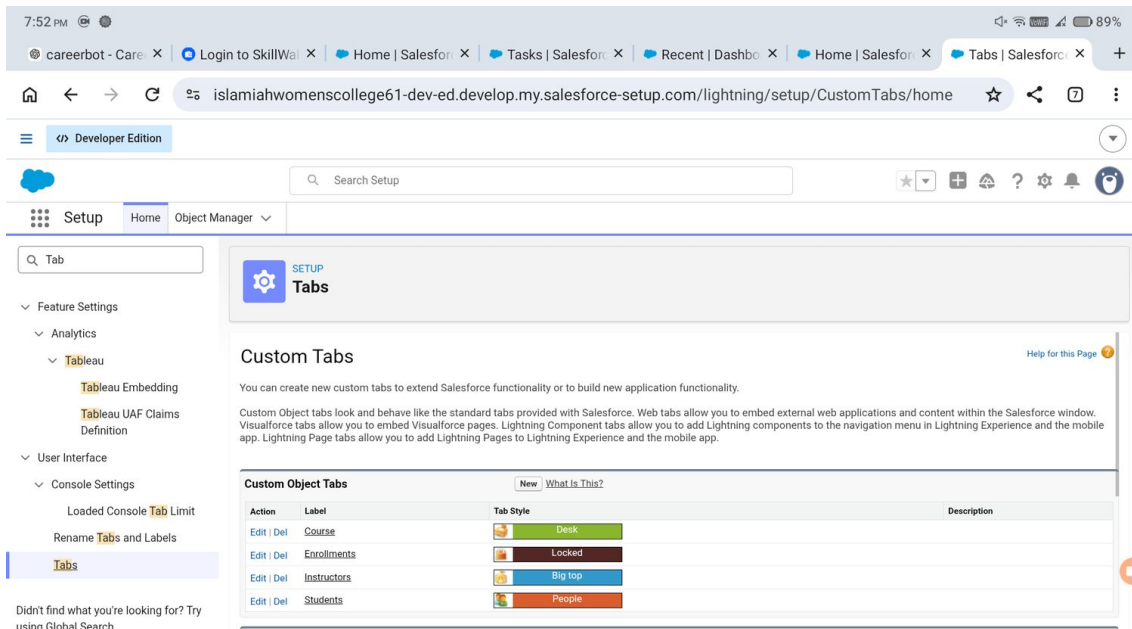
Enrollment fields and lookup relationships

Milestone 3 - Tabs and Lightning App

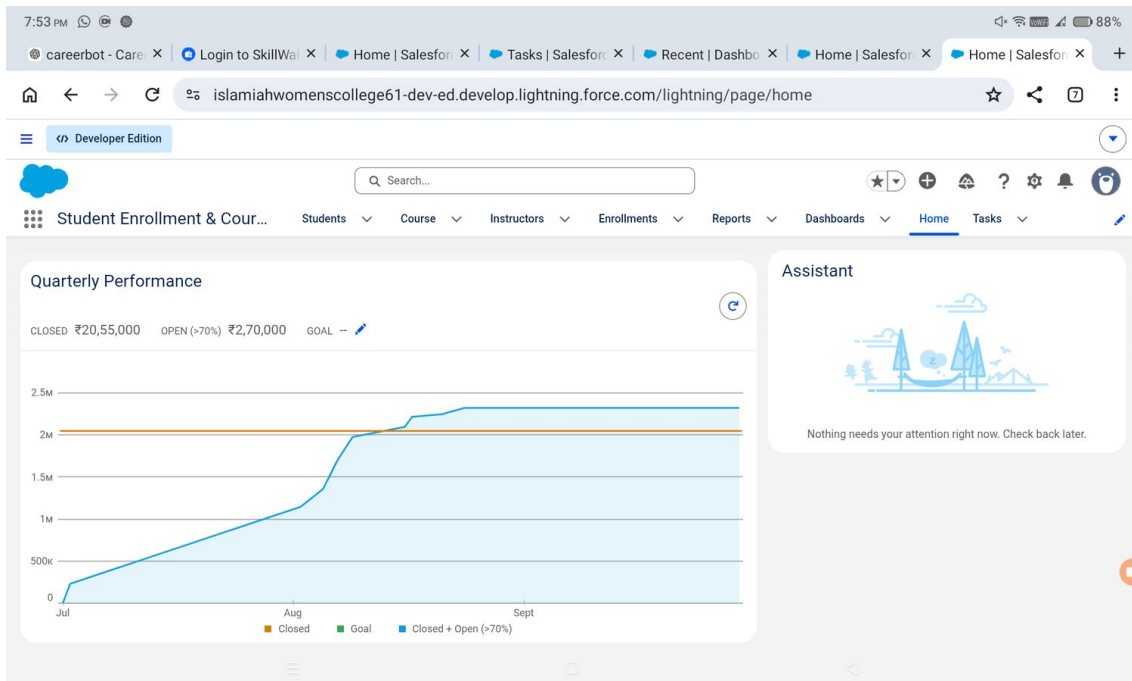
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Tabs and Lightning App

Custom tabs give direct navigation to Student, Course, Instructor and Enrollment. The Student Enrollment & Course Management Lightning App brings these tabs together with Home, Reports, Dashboards and Tasks in one interface.



Custom object tabs



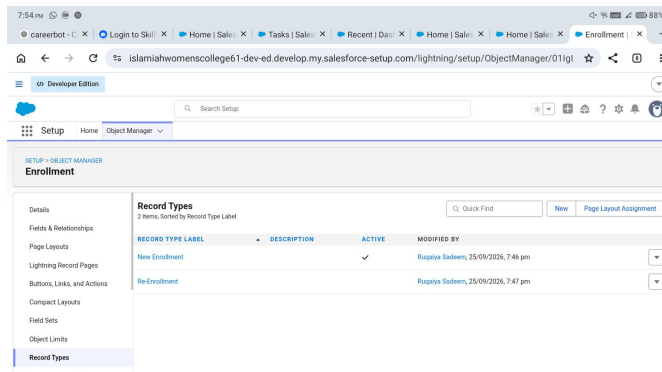
Student Enrollment & Course Management Lightning App

Milestone 4 - Record Types and Page Layouts

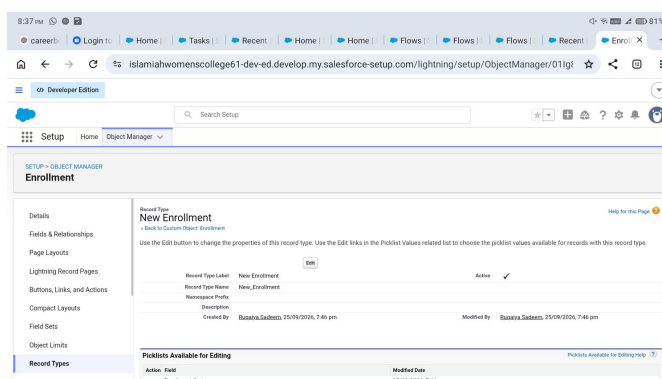
Group ID: SWTID-2026-4095 | Islamiah Women's Arts and Science College | Thiruvalluvar University

Record Types and Page Layouts

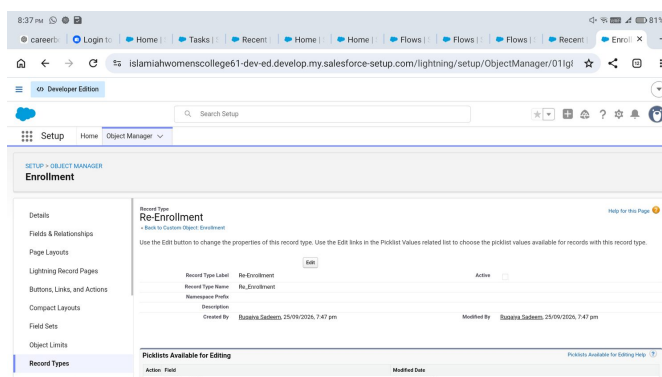
New Enrollment and Re-Enrollment record types separate first-time enrollment from returning-student processing. Each has its own page layout, and related lists show enrollment history on the Student, Course and Instructor pages.



Enrollment record types (New Enrollment, Re-Enrollment)



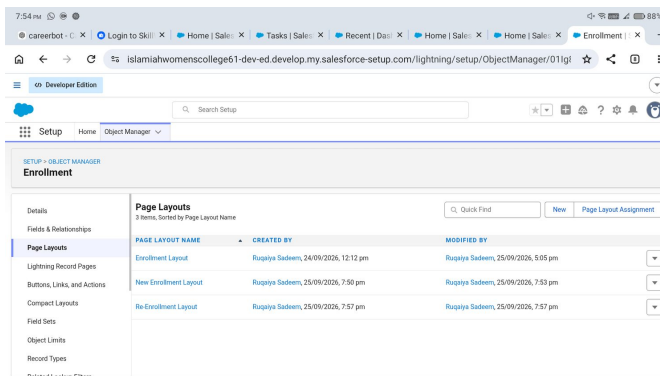
New Enrollment record type



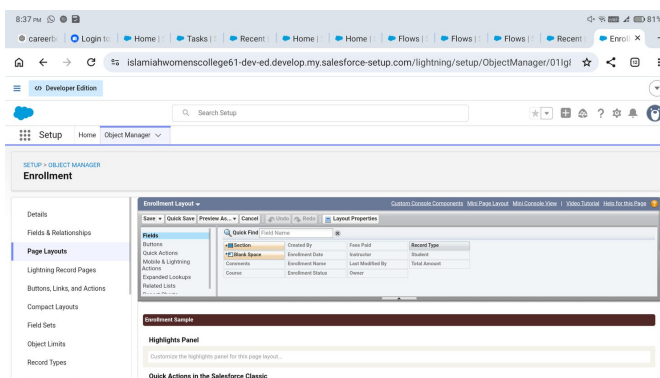
Re-Enrollment record type

Milestone 4 - Page Layouts (continued)

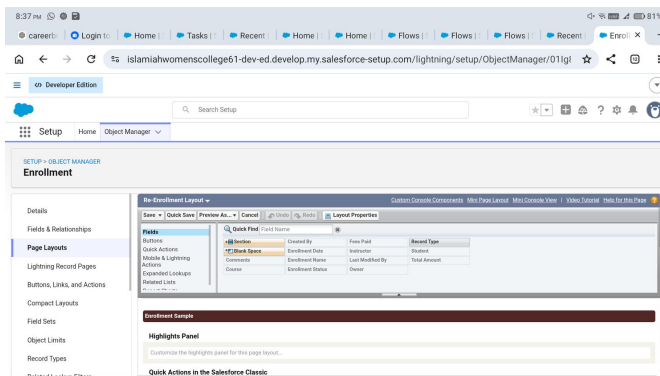
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Enrollment page layouts



Page layout editor (enrollment layout)



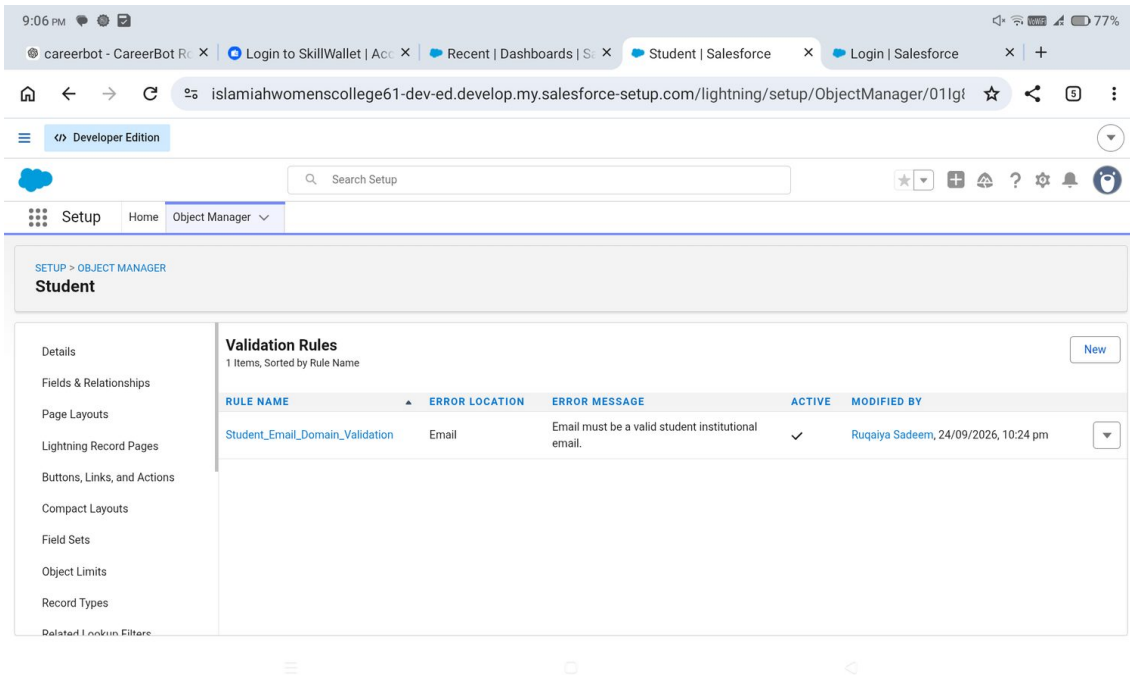
Page layout editor (Re-Enrollment layout)

Milestone 5 - Validation Rules

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Validation Rules

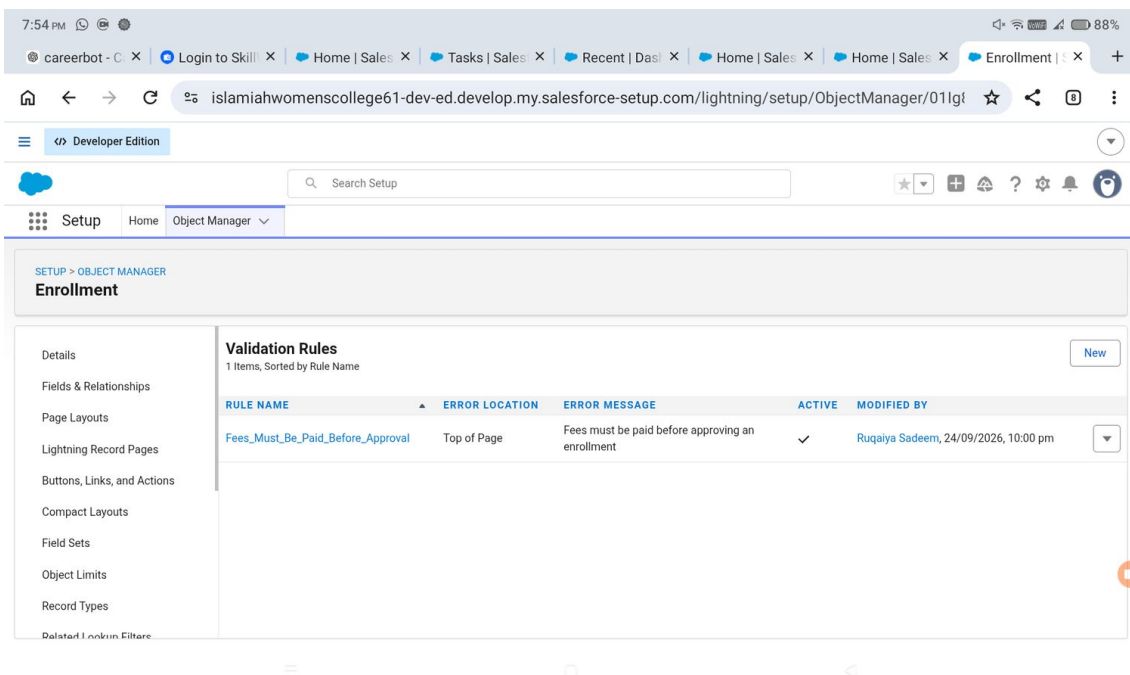
- **Student_Email_Domain_Validation**: checks that the student email meets the configured institutional-domain condition.
- **Fees_Must_Be_Paid_Before_Approval**: prevents an enrollment being marked Approved when Fees Paid is unchecked.



The screenshot shows the Salesforce Setup interface for the 'Student' object. The 'Validation Rules' section is active, displaying a table with one rule: 'Student_Email_Domain_Validation'. The rule is active and was modified by Ruqaiya Sadeem on 24/09/2026 at 10:24 pm. The error message is 'Email must be a valid student institutional email.' and the error location is 'Email'.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Student_Email_Domain_Validation	Email	Email must be a valid student institutional email.	✓	Ruqaiya Sadeem, 24/09/2026, 10:24 pm

Student email validation rule



The screenshot shows the Salesforce Setup interface for the 'Enrollment' object. The 'Validation Rules' section is active, displaying a table with one rule: 'Fees_Must_Be_Paid_Before_Approval'. The rule is active and was modified by Ruqaiya Sadeem on 24/09/2026 at 10:00 pm. The error message is 'Fees must be paid before approving an enrollment' and the error location is 'Top of Page'.

RULE NAME	ERROR LOCATION	ERROR MESSAGE	ACTIVE	MODIFIED BY
Fees_Must_Be_Paid_Before_Approval	Top of Page	Fees must be paid before approving an enrollment	✓	Ruqaiya Sadeem, 24/09/2026, 10:00 pm

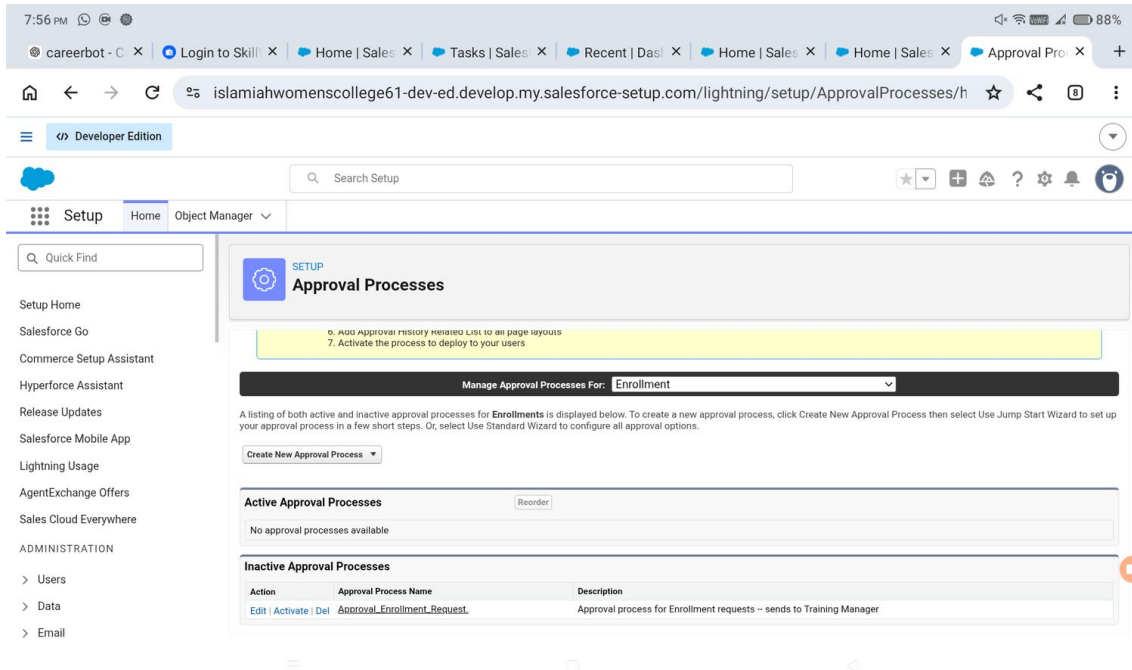
Enrollment fees-paid validation rule

Milestone 5 - Approval Process

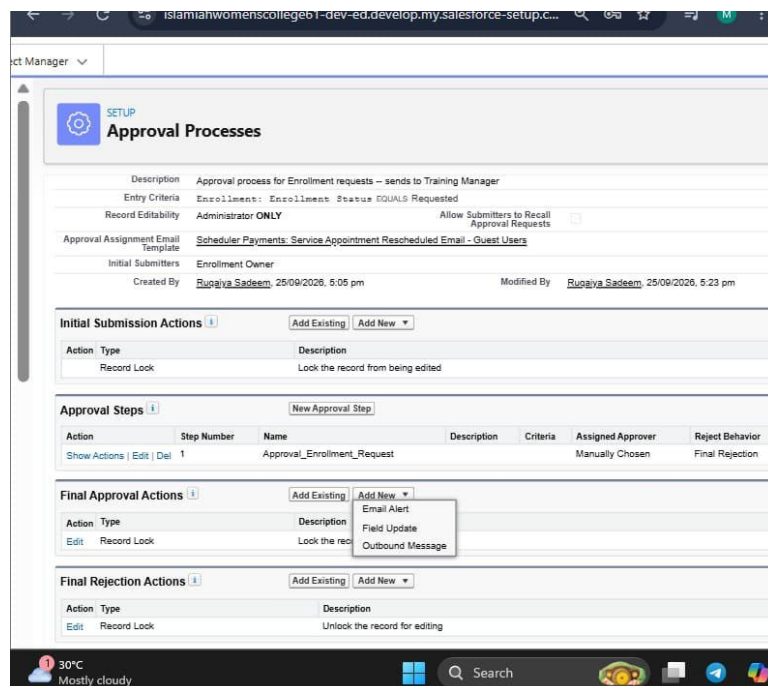
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Approval Process

Approval_Enrollment_Request routes enrollments with Enrollment Status = Requested to the designated approver. Final approval sets Enrollment_Status__c to Approved; final rejection sets the status to Rejected.



Approval processes for Enrollment



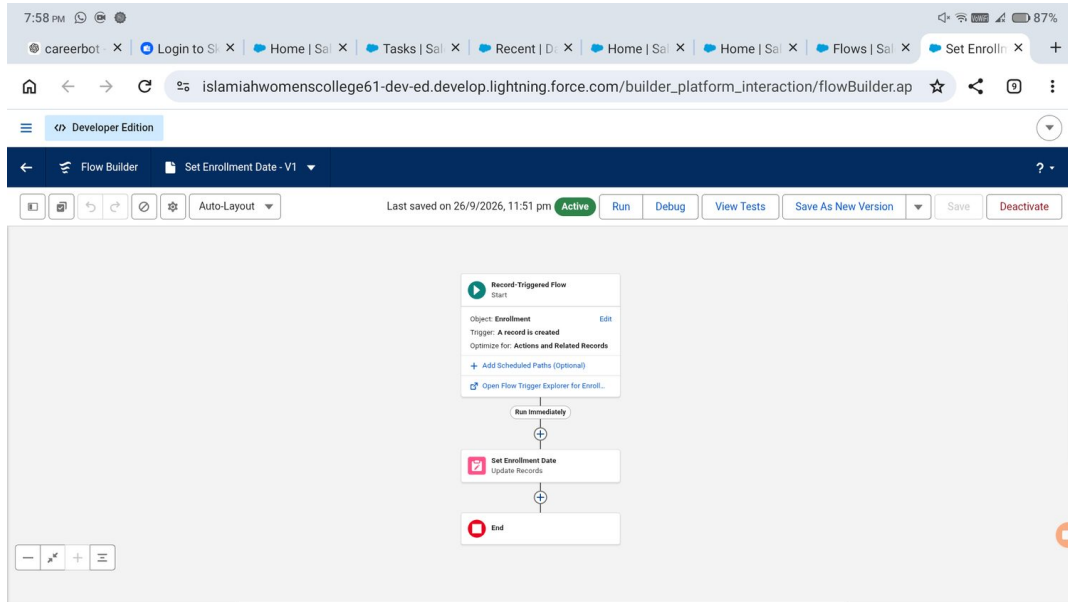
Approval process with final approval / rejection actions

Milestone 6 - Automation Flows

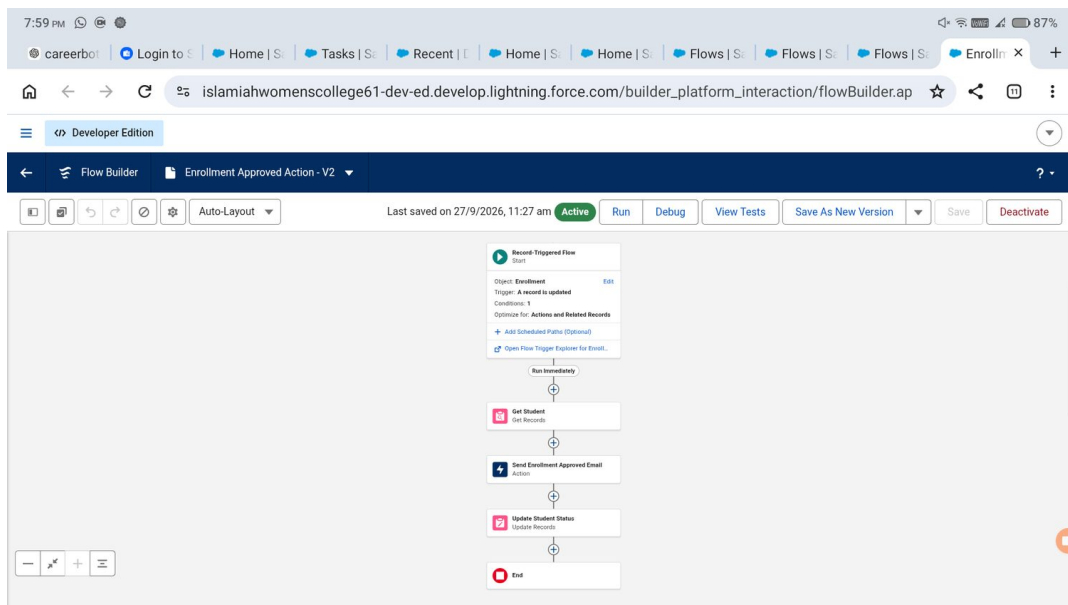
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Automation Flows

- SetEnrollmentDate: sets Enrollment_Date__c to the current date when an enrollment is created.
- Enrollment Approved Action: notifies the student and activates the student after approval.
- AutoAssign_Instructor: assigns an instructor by course category (Technical, Language, Non-Technical).
- Create_Followup_Task: creates a task for the administrator when an enrollment stays in Requested status.



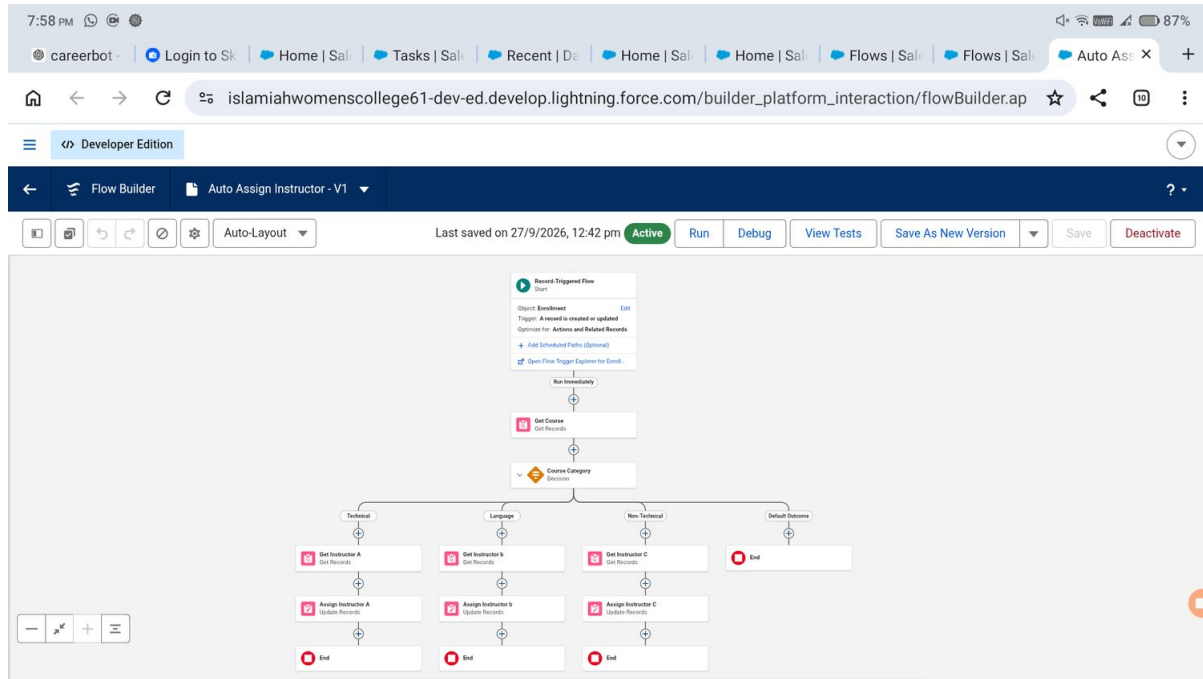
SetEnrollmentDate record-triggered flow



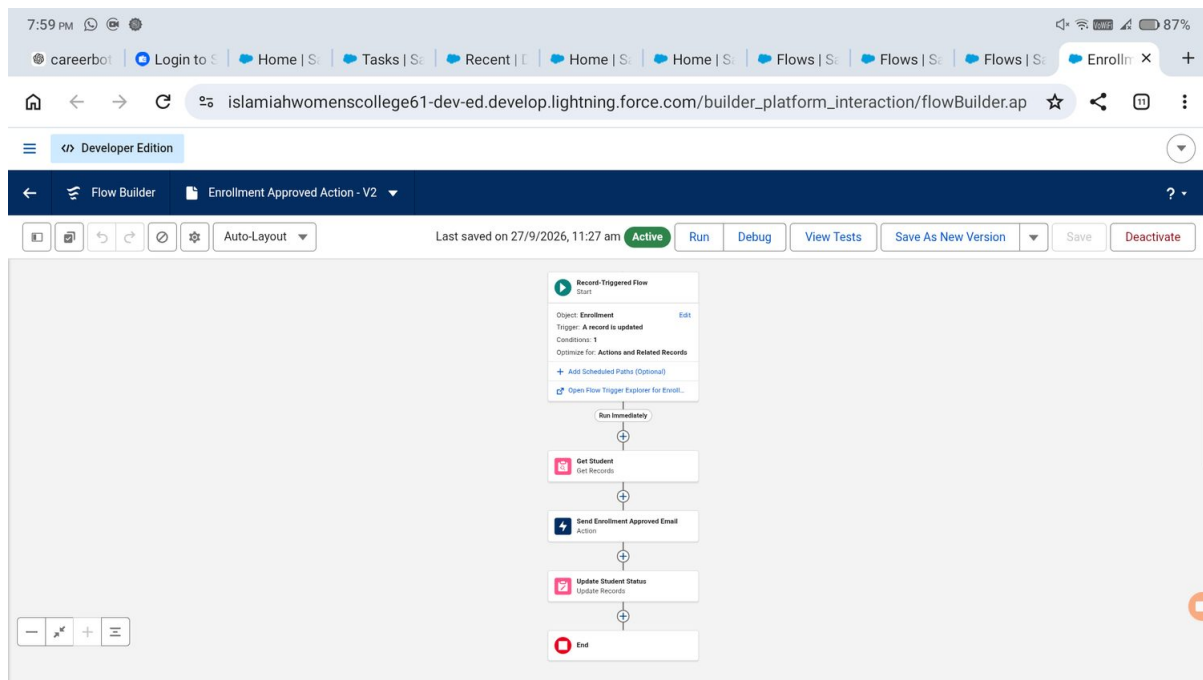
Enrollment Approved Action flow

Milestone 6 - Automation Flows (continued)

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AutoAssign_Instructor flow



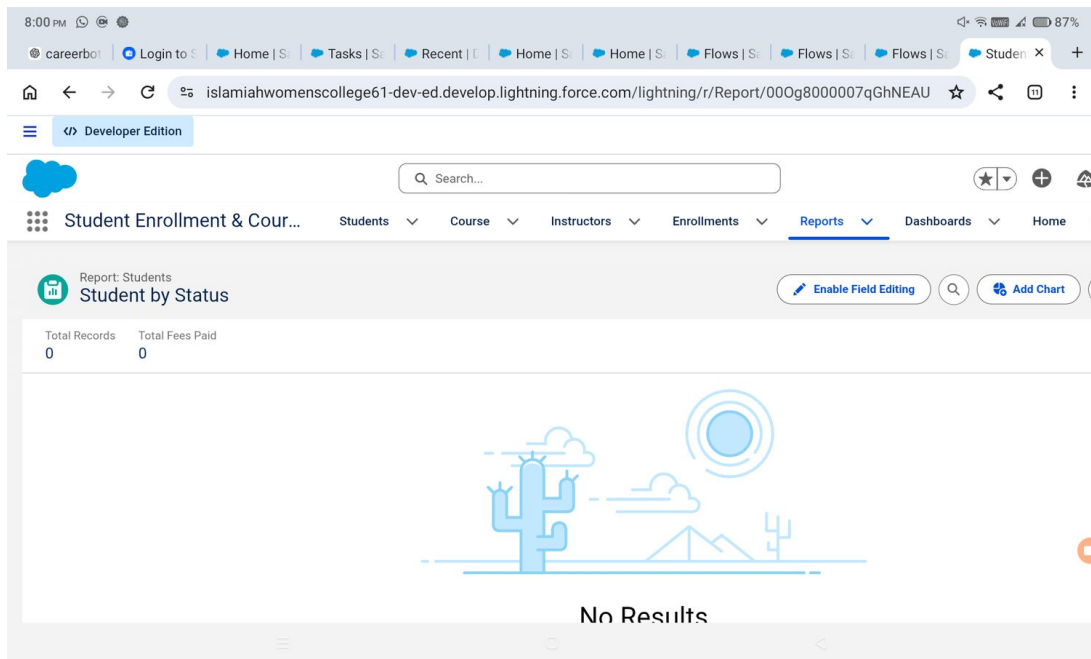
Flow builder screenshot supplied for the follow-up task flow

Milestone 7 - Reports

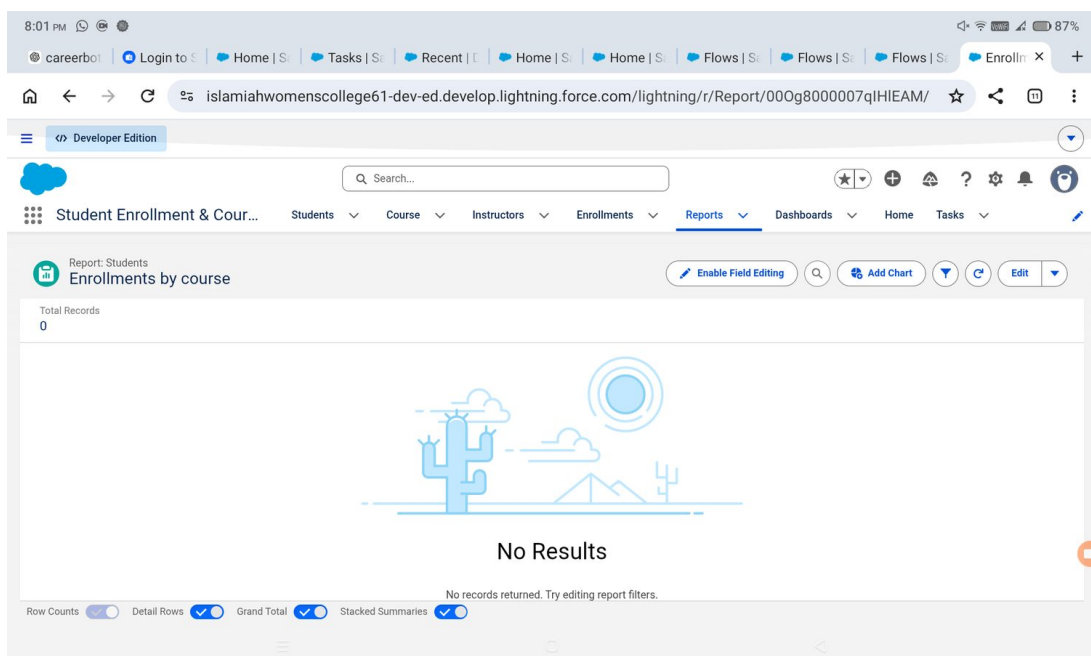
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Reports

- Students by Status - students grouped by Student Status.
- Enrollments by Course - enrollment count per course.
- Pending Enrollment Approvals - enrollments in Requested status.
- Revenue Report - Fees Paid - paid enrollments summarized by course.



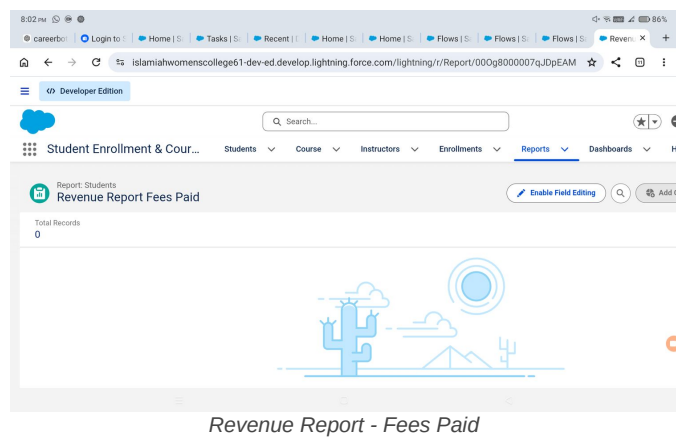
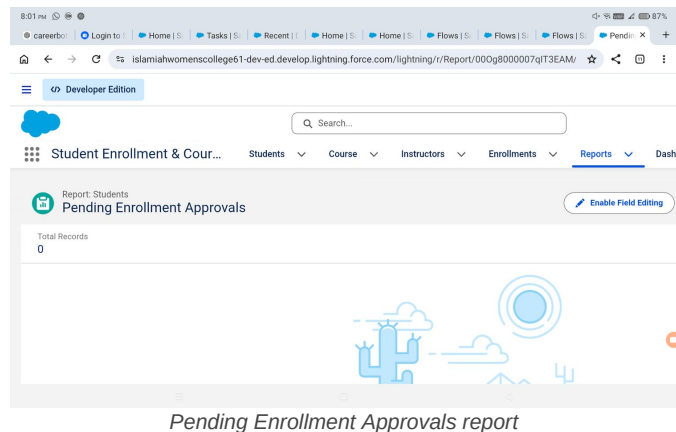
Students by Status report



Enrollments by Course report

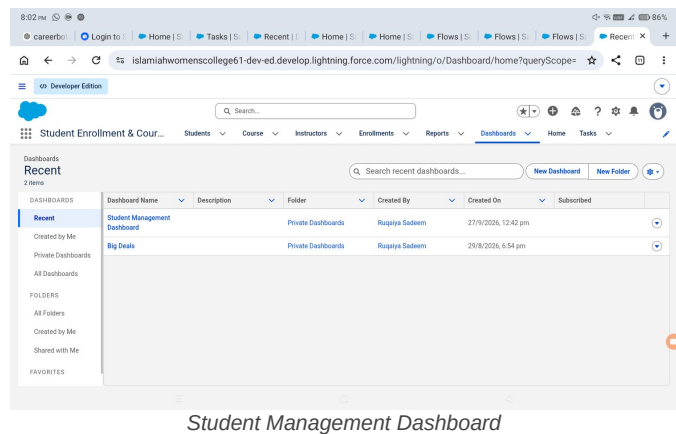
Milestone 7 - Reports and Dashboard (continued)

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Student Management Dashboard

Combines a donut chart (Students by Status), a bar chart (Enrollments by Course), a gauge (Revenue Report) and a table (Pending Enrollment Approvals).

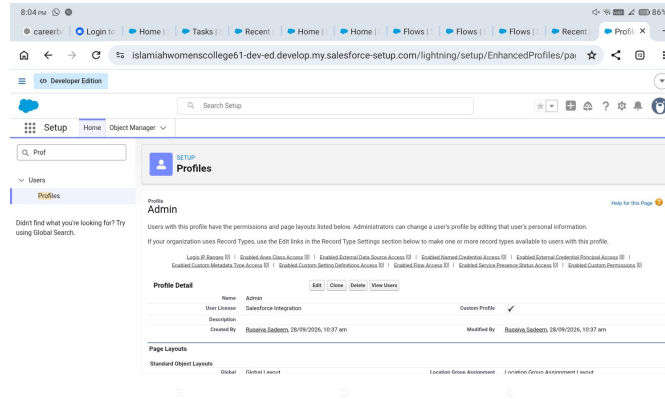


Milestone 8 - Security Configuration

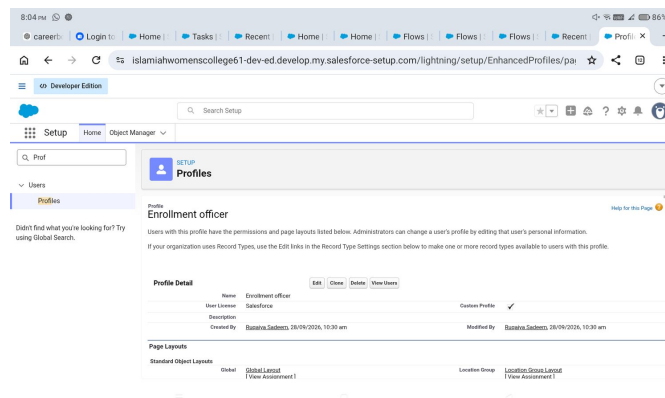
Group ID: SWTID-2026-4095 | Islamiah Women's Arts and Science College | Thiruvalluvar University

Security Configuration

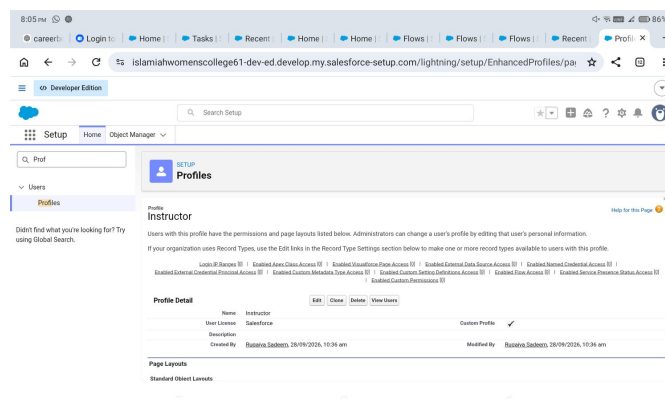
Profiles define baseline access for administrators, enrollment officers and instructors. Permission sets add access where required (for example custom object access and dashboard editing), and sharing settings / OWD control record visibility. The design supports least-privilege access.



Admin profile



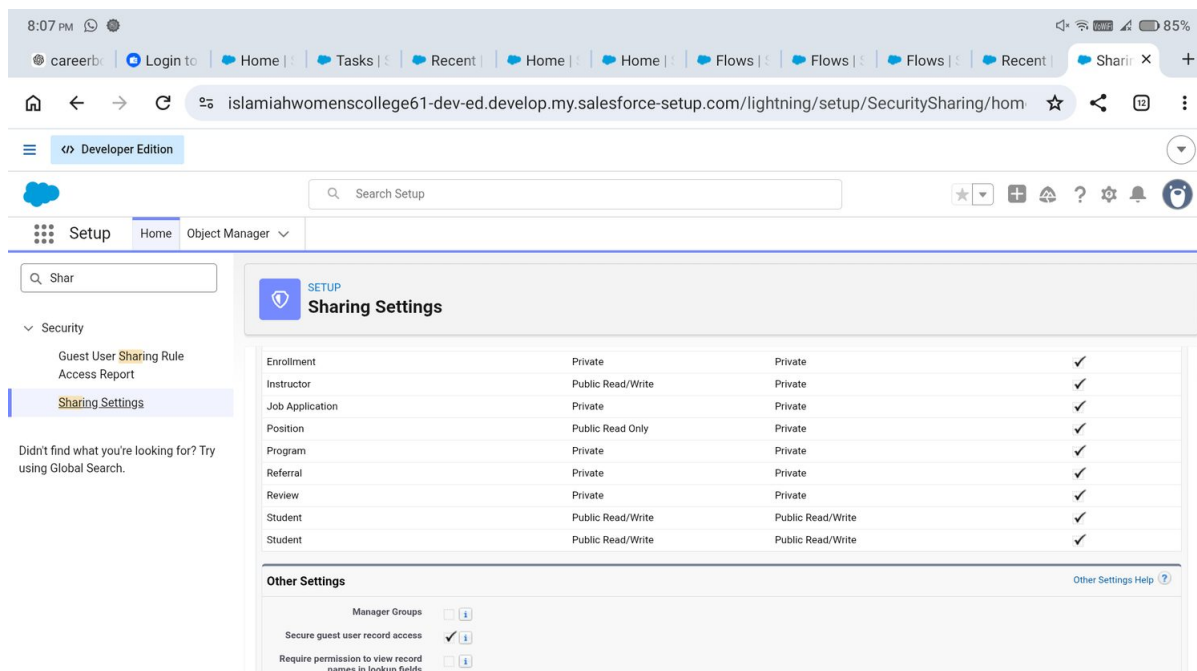
Enrollment Officer profile



Instructor profile

Milestone 8 - Sharing Settings

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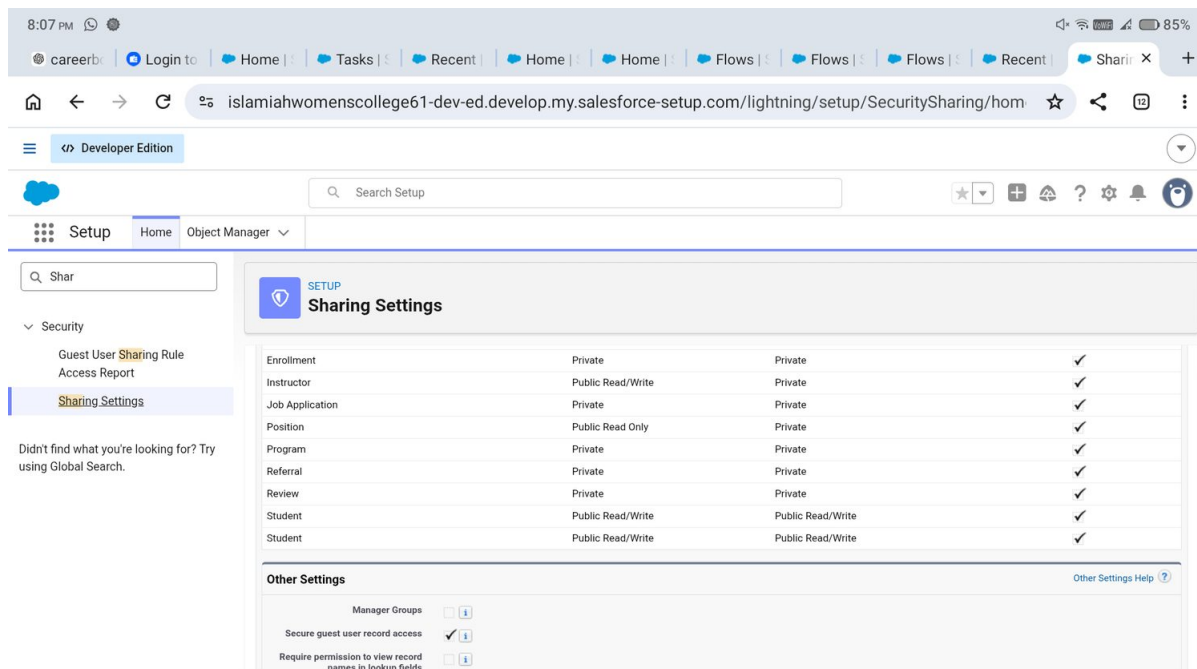
The screenshot shows the Salesforce Setup interface for the 'Sharing Settings' page. The left sidebar contains a search bar with 'Shar' and a list of security-related items, with 'Sharing Settings' highlighted. The main content area displays a table of sharing settings for various objects, with columns for the object name, sharing model, and a checkbox for 'Public Read/Write'. Below the table, there are 'Other Settings' including 'Manager Groups', 'Secure guest user record access', and 'Require permission to view record names in lookup fields'.

Object	Sharing Model	Public Read/Write
Enrollment	Private	Private
Instructor	Public Read/Write	Private
Job Application	Private	Private
Position	Public Read Only	Private
Program	Private	Private
Referral	Private	Private
Review	Private	Private
Student	Public Read/Write	Public Read/Write
Student	Public Read/Write	Public Read/Write

Other Settings

- Manager Groups: ☐
- Secure guest user record access: ☒
- Require permission to view record names in lookup fields: ☐

Sharing settings / organization-wide defaults



This screenshot is identical to the one above, showing the Salesforce Setup interface for the 'Sharing Settings' page. It displays the same table of sharing settings and 'Other Settings' options.

Object	Sharing Model	Public Read/Write
Enrollment	Private	Private
Instructor	Public Read/Write	Private
Job Application	Private	Private
Position	Public Read Only	Private
Program	Private	Private
Referral	Private	Private
Review	Private	Private
Student	Public Read/Write	Public Read/Write
Student	Public Read/Write	Public Read/Write

Other Settings

- Manager Groups: ☐
- Secure guest user record access: ☒
- Require permission to view record names in lookup fields: ☐

Sharing settings / organization-wide defaults (second view)

Testing Strategy and Test Cases

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Testing Strategy

- Functional testing: each feature performs its intended operation.
- Validation testing: invalid or incomplete data is prevented.
- Automation testing: flows and the approval process execute expected actions.
- Reporting and dashboard testing: reports and components show the configured data.
- Security testing: users receive only the intended access.

Representative Test Cases

- TC-01 to TC-10: create Student, Course, Instructor and Enrollment records; check relationships, record types and related lists.
- TC-11 Student email validation - validation message displayed (PASS).
- TC-12 Fees validation - Approved blocked when Fees Paid is false (PASS).
- TC-13 Enrollment date flow - date populated automatically (PASS).
- TC-14 Approval flow - notification / update actions execute (PASS).
- TC-15 Instructor assignment - configured instructor assigned (PASS).
- TC-16 Follow-up task - task created for Requested enrollment (PASS).
- TC-17 to TC-24: approval, rejection, reports and dashboard tests.
- SEC-01 to SEC-04: Admin, Enrollment Officer, Instructor and unauthorized-user access checks.

Full test tables are in the Project Report (chapter 7). Test statuses other than TC-11 to TC-16 are to be confirmed by the group after execution.

Advantages, Limitations, Conclusion and Future Scope

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Advantages

- Centralized academic data management.
- Automated enrollment workflows and structured approvals.
- Automatic instructor assignment by course category.
- Real-time reporting and dashboard visibility.
- Role-based access and cloud-based availability.

Limitations

- Dependence on the Salesforce platform.
- Initial configuration requires planning and administrative effort.
- Future integrations may need additional configuration or external services.

Conclusion

SECMS shows how Salesforce can support the academic enrollment lifecycle in one cloud platform. Centralized data reduces fragmentation, workflows reduce repetitive manual work, validation rules improve data quality, and reports and dashboards improve visibility.

Future Scope

- Phase 1: payment gateway integration.
- Phase 2: student self-service portal.
- Phase 3: advanced analytics.
- Phase 4: mobile access support.